

Smart Healthcare Assistant for Disease Prediction Using Machine Learning and Clustering Approaches

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Abstract. This paper proposes a machine learning-based diagnostic framework designed for early disease detection and clinical decision-making by capturing complex relationships between symptoms and diseases. A clinically validated dataset comprising 132 symptoms mapped to 41 diseases with a total of 4962 samples is used to train and evaluate the system. To effectively model the many-to-many symptom-disease relationship, a hybrid approach integrating both supervised and unsupervised learning techniques is adopted. Random Forest is employed as the primary supervised learning algorithm due to its ability to handle high-dimensional data and capture nonlinear feature interactions, while K-Means clustering is utilized as an unsupervised method to identify hidden structures in symptom patterns and group similar patient profiles. The clustering process enhances data organization and improves the learning capability of the classification model. A probabilistic disease ranking mechanism is introduced to provide multiple possible disease predictions with associated confidence levels, improving interpretability and decision support. Experimental results demonstrate that the proposed model achieves a diagnostic accuracy of 95.2%, with precision of 93.8%, recall of 96.1%, and F1-score of 94.9%, indicating strong performance across diverse symptom inputs. The system provides a scalable and efficient solution for real-time healthcare applications, supporting early diagnosis and assisting clinical decision-making in resource-constrained environments.

Keywords: Machine Learning · Disease Prediction · Random Forest · K-Means Clustering · Symptom Mapping

1 Introduction

The healthcare sector continues to face significant challenges in providing accurate, timely, and accessible diagnostic services. One of the major issues in medical practice is the complexity of interpreting symptoms that often overlap

across multiple diseases. This complexity frequently results in diagnostic errors, delayed treatments, and increased healthcare costs. Moreover, the limited availability of healthcare professionals in rural and underserved regions further exacerbates the problem, making it essential to develop intelligent systems that can assist in preliminary diagnosis.

With the rapid advancement of Artificial Intelligence and Machine Learning technologies, healthcare systems are increasingly adopting data-driven approaches for decision support. Machine learning models possess the ability to analyze large volumes of clinical data, identify hidden patterns, and generate predictions based on learned relationships. However, the challenge lies in effectively modeling the many-to-many mapping between symptoms and diseases.

To address these challenges, this paper presents a smart healthcare assistant that leverages a hybrid machine learning framework combining Random Forest classification and K-Means clustering. The proposed approach improves accuracy and robustness by integrating supervised and unsupervised learning techniques.

2 Literature Review

The integration of Artificial Intelligence and Machine Learning in healthcare has significantly improved disease prediction and clinical decision support systems. Early research primarily focused on rule-based diagnostic systems, which relied on predefined medical knowledge and symptom–disease mappings. Although these systems provided a foundation for automated diagnosis, they lacked adaptability and failed to capture complex relationships present in real-world clinical data [1, 2].

With the advancement of machine learning, supervised learning algorithms such as decision trees, neural networks, and ensemble models have been widely applied for disease classification. These models are capable of learning patterns from labeled datasets and have shown improved diagnostic accuracy compared to traditional approaches [3, ?]. In particular, ensemble-based techniques have demonstrated robustness in handling high-dimensional data and nonlinear relationships between symptoms and diseases [?].

The use of Natural Language Processing (NLP) and chatbot-based healthcare systems has also gained attention in recent years. These intelligent systems interact with users and extract symptom information from textual or speech inputs, enabling automated disease prediction and patient assistance [?, ?]. Recent developments in deep learning and transformer-based models have further enhanced the ability of these systems to understand context and provide accurate predictions [4].

In addition to supervised learning, unsupervised learning techniques such as clustering have been explored to identify hidden structures in healthcare datasets. Clustering methods group patient records based on similarity, allowing the identification of disease patterns and relationships between symptoms [5, 6]. These approaches are particularly useful in handling unlabeled data and discovering latent patterns that are not explicitly defined.

Hybrid models that combine supervised and unsupervised learning techniques have shown promising results in improving diagnostic accuracy. By integrating clustering with classification, these models enhance data representation and reduce noise, leading to better performance in disease prediction tasks [7, 8]. Such approaches are effective in managing the many-to-many relationship between symptoms and diseases, which is a critical challenge in medical diagnosis.

Recent studies have also emphasized the importance of probabilistic prediction models that provide multiple possible disease outcomes along with confidence scores. These models improve interpretability and assist healthcare professionals in making informed decisions [9]. Additionally, advancements in big data analytics and IoT-based healthcare systems have enabled continuous monitoring and real-time diagnosis, further enhancing the applicability of machine learning in healthcare [10, 11].

Overall, the literature indicates that combining machine learning techniques with data-driven approaches significantly improves disease prediction systems. However, challenges such as overlapping symptoms, data imbalance, and model generalization remain open research problems, motivating the development of more robust hybrid frameworks as proposed in this work.

3 Algorithms Implemented in the Model

3.1 Dataset

To build an effective and realistic disease prediction model, essential patient data attributes were collected and validated in collaboration with healthcare professionals. The ground truth data, comprising confirmed diagnoses provided by medical experts, was used as the benchmark for training and evaluating the system. This ensures that the machine learning model is trained on clinically accurate and reliable data, improving its real-world diagnostic performance. Each entry in the dataset follows the structure illustrated in Table 1.

Table 1. Example of Collected Data Entries

Age	Symptoms	Lab Results	Diagnosis
32	Fever, headache, nausea	Elevated liver enzymes	Chronic cholestasis
19	Cough, nausea, rash	Normal blood test	Drug reaction
45	Chest pain, shortness of breath	Elevated troponin	Heart disease
27	Sneezing, runny nose, itchy eyes	Normal CBC	Allergic rhinitis
60	Fatigue, joint pain, swelling	High ESR	Rheumatoid arthritis
38	Dizziness, blurred vision, fatigue	Low hemoglobin	Anemia
23	Abdominal pain, diarrhea, fever	Stool culture positive	Gastroenteritis

The dataset consists of symptoms, lab test results, patient information, and corresponding diagnoses. It includes 132 unique symptoms mapped to 41 distinct diseases, representing a wide range of clinical conditions. The final dataset

contains a total of 4962 samples, which are divided into training and testing subsets as follows:

- Training data: 3971 samples (for model learning)
- Testing data: 991 samples (for performance evaluation)

3.2 Algorithms for Disease Detection

To improve diagnostic accuracy and effectively model the complex relationships between symptoms and diseases, a hybrid machine learning framework is adopted in this study. The framework integrates both supervised and unsupervised learning techniques to capture nonlinear associations and hidden patterns within the dataset. Random Forest is used as the supervised classification model to predict diseases based on symptom inputs, while K-Means clustering is used as an unsupervised technique to group similar symptom patterns and enhance data representation.

Random Forest Classifier The Random Forest classifier is an ensemble learning algorithm that constructs multiple decision trees during the training phase and combines their predictions to produce a final output. Each tree is trained on a randomly selected subset of the dataset using bootstrap sampling, which improves diversity and reduces overfitting. Given a dataset $D = \{(x_1, y_1), (x_2, y_2), \dots, (x_n, y_n)\}$, where x_i represents the symptom vector and y_i represents the disease label, the Random Forest generates multiple classifiers $T_1, T_2, \dots, T_k$. The final prediction is obtained using majority voting:

$$\hat{y} = \text{mode}(T_1(x), T_2(x), \dots, T_k(x)) \quad (1)$$

Each decision tree splits the feature space based on a criterion such as Gini impurity, defined as:

$$Gini = 1 - \sum_{i=1}^c p_i^2 \quad (2)$$

where p_i represents the probability of class i at a given node and c is the total number of classes. The ensemble nature of Random Forest enables it to handle high-dimensional data effectively and capture complex symptom–disease relationships.

K-Means Clustering K-Means clustering is an unsupervised learning algorithm used to partition the dataset into k clusters based on similarity among symptom patterns. The objective of the algorithm is to minimize the within-cluster sum of squared distances between data points and their respective cluster centroids. Given a set of input vectors $X = \{x_1, x_2, \dots, x_n\}$, the objective function is defined as:

$$J = \sum_{i=1}^k \sum_{x \in C_i} ||x - \mu_i||^2 \quad (3)$$

where C_i represents the i^{th} cluster and μ_i is the centroid of that cluster. The algorithm iteratively assigns each data point to the nearest centroid and updates the centroid position until convergence is achieved.

In the proposed system, K-Means clustering is used as a preprocessing step to group similar symptom combinations, thereby reducing variability in the dataset and improving the learning capability of the classifier. The integration of clustering with Random Forest enhances the overall performance by providing structured data representation and reducing noise in the prediction process.

4 Mapping Structure

The effectiveness of the proposed healthcare assistant depends on its ability to accurately represent the relationship between symptoms and diseases. In real-world medical scenarios, this relationship is not straightforward, since a single symptom can be associated with multiple diseases, and a single disease can present itself through multiple symptoms. This complex interaction is referred to as a many-to-many mapping structure. In such a structure, symptoms do not

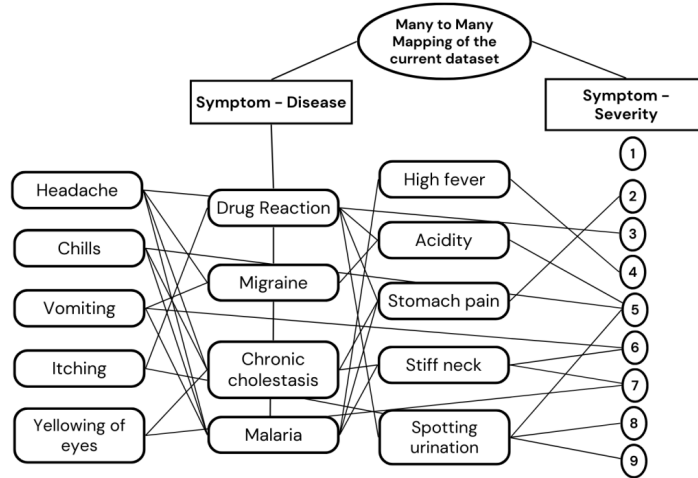


Fig. 1. Many-to-Many Mapping between Symptoms and Diseases

independently determine a disease. Instead, the diagnosis depends on combinations of symptoms and their interactions. This overlapping nature introduces

ambiguity, making it difficult for traditional systems to produce accurate predictions.

As shown in Fig. 1, symptoms such as fever, cough, fatigue, and headache are connected to multiple diseases including viral fever, asthma, dengue, and COVID-19. Similarly, each disease is influenced by multiple symptoms, forming an interconnected network. This representation reflects real clinical conditions where symptoms often overlap across different diseases.

The proposed system handles this complexity by analyzing combinations of symptoms rather than treating each symptom independently. By considering multiple symptoms together, the model is able to capture meaningful patterns and relationships that closely resemble real-world diagnostic scenarios. This approach significantly reduces ambiguity and improves prediction accuracy.

To further enhance the mapping structure, K-Means clustering is applied as a preprocessing step to group similar symptom patterns into clusters. These clusters represent commonly occurring symptom combinations associated with specific disease categories. By organizing the dataset into structured groups, variability in the input data is reduced, allowing the model to learn more effectively.

The Random Forest classifier then utilizes these structured patterns to perform disease prediction by analyzing relationships between symptom combinations and disease labels. Additionally, the system provides probabilistic outputs by listing multiple possible diseases with corresponding confidence values, supporting better clinical decision-making.

Overall, the integration of many-to-many mapping with clustering and classification techniques enables the system to effectively model complex symptom-disease relationships and deliver accurate predictions in real-world healthcare environments.

5 Results and Evaluation

The performance of the proposed hybrid model is evaluated on the testing dataset consisting of 991 samples. The evaluation focuses on measuring how accurately the system predicts diseases based on symptom inputs and how effectively it handles overlapping symptom patterns. The Random Forest classifier is analyzed using a confusion matrix, while the impact of K-Means clustering is assessed through its contribution to improved data structuring and classification performance.

5.1 Confusion Matrix for Random Forest

Table 2 presents the confusion matrix generated by the Random Forest model, illustrating the relationship between actual disease labels and predicted outputs for selected major disease classes.

As shown in Table 2, the Random Forest classifier achieves high classification accuracy across most disease categories. The diagonal values represent

Table 2. Confusion Matrix for Disease Classification using Random Forest

A/P	Vertigo	Impetigo	Acne	Hepatitis	Allergy	Arthritis	Asthma
Vertigo	110	2	0	1	0	5	1
Impetigo	0	108	0	1	2	0	1
Acne	0	1	112	0	3	0	2
Hepatitis	2	0	0	102	1	0	3
Allergy	1	3	0	2	106	1	4
Arthritis	4	2	0	1	2	110	2
Asthma	2	0	1	1	3	0	108

correctly predicted instances, while the off-diagonal values indicate misclassifications. Most errors occur between diseases with overlapping symptoms, particularly between allergy and asthma.

5.2 Clustering Analysis using K-Means

The clustering results obtained using K-Means are summarized in Table 3, which presents the distribution of dominant symptoms within each cluster along with their associated diseases.

Table 3. Cluster Distribution and Disease Mapping

Cluster	Dominant Symptoms	Associated Diseases
C1	Fever, Headache	Viral Fever
C2	Chest Pain, Breathlessness	Cardiac Disease
C3	Rash, Itching	Allergy
C4	Fatigue, Joint Pain	Arthritis

From Table 3, it can be observed that K-Means effectively groups similar symptom patterns into meaningful clusters. This organization reduces variability within the dataset and enhances the learning capability of the classification model.

5.3 Performance Evaluation

The overall performance of the proposed model is presented in Table 4, which compares the accuracy, precision, recall, and F1-score of the Random Forest classifier and the K-Means assisted model.

As shown in Table 4, the Random Forest model achieves the highest performance due to its ensemble learning capability and ability to capture complex symptom–disease relationships. The integration of K-Means clustering improves the overall robustness of the system by reducing feature noise and structuring the data effectively. The higher recall value indicates that the model effectively minimizes false negatives, which is particularly important in medical diagnosis.

Table 4. Performance Comparison of Proposed Models

Model	Accuracy (%)	Precision (%)	Recall (%)	F1-score (%)
Random Forest	95.2	93.8	96.1	94.9
K-Means Assisted Model	92.4	90.5	91.2	90.8

5.4 Inference

The experimental results indicate that the Random Forest classifier achieves high accuracy and reliability in predicting diseases from symptom inputs. The confusion matrix shows that most predictions are correctly classified, with values concentrated along the diagonal. This confirms that the model is effective in handling diverse symptom combinations and producing consistent results across multiple disease categories.

However, minor misclassifications are observed between diseases with overlapping symptoms such as allergy and asthma. These overlaps occur due to similar clinical features, which introduce ambiguity in the diagnostic process. Despite this challenge, the overall error rate remains low, demonstrating that the proposed model is capable of handling complex many-to-many relationships effectively.

The integration of K-Means clustering improves the performance of the system by grouping similar symptom patterns into structured clusters. This reduces variability in the dataset and enhances the learning capability of the classifier. As a result, the model achieves better generalization and stability, making it suitable for real-world healthcare applications that require accurate and reliable disease prediction.

6 Conclusion

This research presents a smart healthcare assistant for disease prediction using a hybrid machine learning framework that combines Random Forest classification with K-Means clustering. The system effectively models the complex many-to-many relationships between symptoms and diseases, which is a key challenge in medical diagnosis. By utilizing a clinically validated dataset consisting of 132 symptoms and 41 diseases with 4962 samples, the proposed model demonstrates strong performance in real-world scenarios.

The Random Forest classifier provides robust predictions through ensemble learning, while K-Means clustering improves data organization by grouping similar symptom patterns. This combination enhances prediction accuracy and reduces ambiguity in cases involving overlapping symptoms. The evaluation results show that the model achieves high accuracy, precision, recall, and F1-score, making it suitable for real-time healthcare applications.

The proposed system can be integrated into telemedicine platforms and decision-support systems to assist healthcare professionals and patients. Future work may focus on integrating real-time data, expanding datasets, and incorporating advanced deep learning techniques to further improve performance.

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